COMPLETE FINANCE + ACCOUNTING KNOWLEDGE FOR MBB CASE INTERVIEWS

EASY DEFINITIONS + FORMULAS + EXAMPLES + MBB APPLICATIONS

PART 1 — BASIC FINANCE AND ACCOUNTING

1. ACCOUNTING

Easy definition

Accounting is the process of **recording, organizing, and reporting a company's financial activities**.

It tells us:

- How much the company earned
- How much it spent
- How much profit it made
- What it owns
- What it owes
- How much cash it has

Example

A company:

- Sells products worth ₹100 crore
- Spends ₹80 crore
- Makes ₹20 crore profit

Accounting records these activities and shows the company's financial performance.

MBB use

In a case interview, accounting helps analyze:

- Why profit decreased
- Which costs increased
- Whether a company is financially healthy
- Whether an acquisition target is attractive

---

2. FINANCE

Easy definition

Finance is the study of **how money is raised, invested, managed, and used to create value**.

Finance answers:

- Should the company invest?
- Should the company borrow money?
- Is an acquisition worth buying?

- What return will an investment generate?
- How much is a company worth?

Example

A company must choose between:

- Investing ₹100 crore in a new factory
- Acquiring another company for ₹100 crore

Finance helps determine which option creates more value.

MBB use

Finance is used in:

- M&A cases
- Investment decisions
- Market-entry cases
- Growth cases
- Business valuation

---

3. ACCOUNTING VS FINANCE

Accounting	Finance
Records financial activity	Makes financial decisions
Focuses on past and current performance	Focuses on future value
Reports revenue, costs, assets, and profit	Evaluates investments and returns
Answers: “What happened?”	Answers: “What should we do?”

PART 2 — THE THREE FINANCIAL STATEMENTS

4. FINANCIAL STATEMENTS

Easy definition

Financial statements are formal reports showing a company’s:

- Performance
- Financial position
- Cash movement

The three main financial statements are:

1. Income Statement
2. Balance Sheet
3. Cash Flow Statement

---

5. INCOME STATEMENT

Easy definition

The Income Statement shows:

How much money the company earned, how much it spent, and how much profit it made during a period.

It covers a period such as:

- One month
- One quarter
- One year

Basic structure

Revenue

− Cost of Goods Sold

= Gross Profit

− Operating Expenses

= EBITDA

− Depreciation and Amortization

= EBIT

− Interest

= Profit Before Tax

− Tax

= Net Income

---

6. REVENUE

Easy definition

Revenue is the **total money earned from selling products or services before subtracting costs**.

Revenue is also called:

- Sales
- Sales revenue
- Turnover

Formula

Revenue = Selling Price × Quantity Sold

Example

Selling price:

₹1,000 per product

Products sold:

100,000

Revenue:

₹1,000 × 100,000

= **₹10 crore**

Important

Revenue is **not profit**.

A company can have high revenue but low profit if costs are high.

MBB use

Revenue analysis is used in:

- Profitability cases
- Pricing cases
- Growth cases
- Market-entry cases

---

7. COST OF GOODS SOLD — COGS

Easy definition

COGS is the **direct cost of making or buying the products that the company sells**.

Examples

For a manufacturing company:

- Raw materials
- Direct factory labor
- Production costs

For a retailer:

- Cost paid to suppliers for products

For a restaurant:

- Food ingredients
- Packaging

Formula

Gross Profit = Revenue − COGS

Example

Revenue:

₹100 crore

COGS:

₹60 crore

Gross profit:

₹40 crore

MBB use

If gross profit decreases, investigate:

- Higher raw-material prices
- Higher production costs
- Lower selling prices
- Changes in product mix

---

8. GROSS PROFIT

Easy definition

Gross profit is the money left after paying the **direct cost of producing or buying the products sold**.

Formula

Gross Profit = Revenue – COGS

Example

Revenue:

₹100 crore

COGS:

₹60 crore

Gross profit:

₹40 crore

---

9. GROSS PROFIT MARGIN

Easy definition

Gross margin shows:

How much gross profit the company keeps from every ₹100 of revenue.

Formula

Gross Margin = Gross Profit ÷ Revenue × 100

Example

Gross profit:

₹40 crore

Revenue:

₹100 crore

Gross margin:

40%

Meaning

For every ₹100 of sales:

- ₹60 goes to direct product costs
- ₹40 remains to pay operating expenses and generate profit

---

10. OPERATING EXPENSES — OPEX

Easy definition

Operating expenses are the costs required to **run the company's normal business operations**.

Examples

- Employee salaries
- Marketing
- Rent
- Administration
- IT costs
- Office expenses
- Research and development

Important

OPEX is different from COGS.

COGS:

Direct cost of making or buying products.

OPEX:

Cost of running the business.

---

11. EBITDA

Full form

Earnings Before Interest, Taxes, Depreciation, and Amortization

Easy definition

EBITDA shows the company's **operating earnings before considering financing, taxes, and certain accounting expenses**.

It is often used to compare the operating performance of companies.

Formula

EBITDA = Revenue – COGS – Operating Expenses

or:

EBITDA = EBIT + Depreciation + Amortization

Example

Revenue:

₹100 crore

COGS:

₹60 crore

Operating expenses:

₹15 crore

EBITDA:

₹25 crore

Meaning

The company generated ₹25 crore from its operations before:

- Interest
- Tax
- Depreciation
- Amortization

MBB use

EBITDA is important in:

- M&A cases
- Company valuation
- Profitability analysis
- Comparing companies

---

12. EBITDA MARGIN

Easy definition

EBITDA margin shows how much operating earnings the company generates from every ₹100 of revenue.

Formula

EBITDA Margin = EBITDA ÷ Revenue × 100

Example

EBITDA:

₹25 crore

Revenue:

₹100 crore

EBITDA margin:

25%

Meaning

The company earns ₹25 in EBITDA from every ₹100 of revenue.

---

13. DEPRECIATION

Easy definition

Depreciation is the **accounting reduction in the value of a physical asset over its useful life**.

Physical assets include:

- Machinery
- Factories
- Vehicles
- Equipment

Example

A machine costs:

₹100 lakh

Useful life:

10 years

Annual depreciation:

₹100 lakh ÷ 10

= ₹10 lakh

Important

The company paid ₹100 lakh when it bought the machine.

It does not pay ₹10 lakh cash every year.

The ₹10 lakh is an accounting expense that spreads the machine's cost across its useful life.

MBB use

Depreciation:

- Reduces accounting profit
- Does not directly reduce cash in the year it is recorded
- Is added back when calculating cash flow

---

14. AMORTIZATION

Easy definition

Amortization is similar to depreciation but usually applies to **intangible assets**.

Intangible assets include:

- Patents
- Software
- Licenses
- Copyrights

Example

A company buys a software license for:

₹50 lakh

Useful life:

5 years

Annual amortization:

₹10 lakh

---

15. EBIT

Full form

Earnings Before Interest and Taxes

Easy definition

EBIT is the company's **operating profit after subtracting operating expenses, depreciation, and amortization but before interest and tax**.

EBIT is also called:

- Operating profit
- Operating income

Formula

EBIT = EBITDA − Depreciation − Amortization

Example

EBITDA:

₹25 crore

Depreciation:

₹4 crore

Amortization:

₹1 crore

EBIT:

₹20 crore

---

16. EBIT MARGIN

Formula

EBIT Margin = EBIT ÷ Revenue × 100

Example

EBIT:

₹20 crore

Revenue:

₹100 crore

EBIT margin:

20%

Meaning

The company generates ₹20 of operating profit for every ₹100 of revenue.

---

17. INTEREST EXPENSE

Easy definition

Interest expense is the **cost paid by a company for borrowing money**.

Example

Debt:

₹100 crore

Interest rate:

10%

Annual interest:

₹10 crore

MBB use

High interest expense may indicate:

- High debt
- Financial risk
- Lower net profit

---

18. PROFIT BEFORE TAX — PBT

Easy definition

PBT is the profit remaining after paying operating expenses and interest but before paying tax.

Formula

PBT = EBIT − Interest Expense

Example

EBIT:

₹20 crore

Interest:

₹5 crore

PBT:

₹15 crore

---

19. TAX

Easy definition

Tax is the amount paid by a company to the government based on taxable profit.

Formula

Tax = Profit Before Tax × Tax Rate

Example

PBT:

₹15 crore

Tax rate:

30%

Tax:

₹4.5 crore

---

20. NET INCOME

Easy definition

Net income is the **final accounting profit remaining after all expenses, interest, and taxes are deducted**.

It is also called:

- Net profit
- Profit after tax
- Bottom-line profit

Formula

Net Income = PBT − Tax

Example

PBT:

₹15 crore

Tax:

₹4.5 crore

Net income:

₹10.5 crore

---

21. COMPLETE INCOME-STATEMENT EXAMPLE

Item	Amount
Revenue	₹100 crore
COGS	−₹60 crore
Gross Profit	₹40 crore
Operating Expenses	−₹15 crore
EBITDA	₹25 crore
Depreciation	−₹4 crore
Amortization	−₹1 crore
EBIT	₹20 crore
Interest	−₹5 crore
Profit Before Tax	₹15 crore
Tax	−₹4.5 crore
Net Income	**₹10.5 crore**

PART 3 — BALANCE SHEET

22. BALANCE SHEET

Easy definition

The Balance Sheet shows:

What the company owns, what it owes, and how much value belongs to its owners at a specific date.

Main equation:

Assets = Liabilities + Shareholders' Equity

---

23. ASSETS

Easy definition

Assets are **valuable resources owned or controlled by a company that can provide future benefits**.

Examples

- Cash

- Inventory
- Machinery
- Factories
- Buildings
- Patents

---

24. CURRENT ASSETS

Easy definition

Current assets are assets expected to be used, sold, or converted into cash within one year.

Examples

- Cash
- Inventory
- Accounts receivable

---

25. NON-CURRENT ASSETS

Easy definition

Non-current assets are long-term assets used for more than one year.

Examples

- Factories
- Machinery
- Buildings
- Long-term investments

---

26. LIABILITIES

Easy definition

Liabilities are **money or obligations that the company owes to other parties**.

Examples

- Bank loans
- Supplier payments
- Bonds
- Taxes payable

---

27. CURRENT LIABILITIES

Easy definition

Current liabilities are obligations due within one year.

Examples

- Accounts payable
- Short-term loans
- Unpaid expenses

---

28. LONG-TERM LIABILITIES

Easy definition

Long-term liabilities are obligations due after one year.

Examples

- Long-term debt
- Bonds
- Long-term loans

---

29. SHAREHOLDERS' EQUITY

Easy definition

Shareholders' equity is the value remaining for the owners after all liabilities are subtracted from assets.

Formula

Equity = Assets − Liabilities

Example

Assets:

₹500 crore

Liabilities:

₹300 crore

Equity:

₹200 crore

---

30. BALANCE-SHEET EXAMPLE

Assets	**Amount**
Cash	₹50 crore
Inventory	₹70 crore
Accounts receivable	₹80 crore
Factory and machinery	₹300 crore

Assets	**Amount**
Total Assets	**₹500 crore**

Liabilities and Equity	**Amount**
Accounts payable	₹60 crore
Debt	₹240 crore
Shareholders' equity	₹200 crore
Total	**₹500 crore**

The balance sheet balances:

₹500 crore = ₹500 crore

PART 4 — CASH FLOW

31. CASH FLOW

Easy definition

Cash flow means the **actual movement of money into and out of a company**.

Cash inflow:

Money enters the company.

Cash outflow:

Money leaves the company.

---

32. CASH FLOW STATEMENT

Easy definition

The Cash Flow Statement shows:

Where the company received cash and where it used cash during a period.

Three parts:

1. Operating cash flow
2. Investing cash flow
3. Financing cash flow

---

33. CASH FLOW FROM OPERATIONS — CFO

Easy definition

Cash generated or used by the company's normal business activities.

Examples

Cash received from:

- Customers

Cash paid to:

- Suppliers
- Employees
- Operating expenses

---

34. CASH FLOW FROM INVESTING — CFI

Easy definition

Cash used to buy or received from selling long-term assets or investments.

Examples

- Buying machinery
- Building a factory
- Buying another company
- Selling a business

---

35. CASH FLOW FROM FINANCING — CFF

Easy definition

Cash received from or paid to lenders and owners.

Examples

- Borrowing money
- Repaying loans
- Issuing shares
- Paying dividends

---

36. NET CHANGE IN CASH

Formula

Net Change in Cash = CFO + CFI + CFF

Example

Operating cash flow:

₹100 crore

Investing cash flow:

−₹70 crore

Financing cash flow:

₹20 crore

Net change:

₹50 crore

---

37. PROFIT VS CASH FLOW

Easy explanation

Profit is an accounting measure.

Cash flow is actual cash movement.

They may be different.

Example

A company sells products worth ₹100 crore on credit.

The company records:

Revenue:

₹100 crore

But customers pay only:

₹40 crore

Therefore:

Profit may be high, but cash received is low.

MBB importance

A company can be profitable but still face a cash shortage.

PART 5 — COSTS AND PROFITABILITY

38. FIXED COST

Easy definition

A fixed cost is a cost that **does not change much when production or sales volume changes in the short term**.

Examples

- Factory rent
- Office rent
- Salaried employees
- Insurance

Example

Annual factory rent:

₹10 crore

Whether the company produces:

100,000 units or 200,000 units

Rent remains approximately:

₹10 crore

---

39. VARIABLE COST

Easy definition

A variable cost is a cost that **changes when the number of units produced or sold changes**.

Examples

- Raw materials
- Packaging
- Shipping per unit
- Sales commission

Formula

Total Variable Cost = Variable Cost per Unit × Quantity

---

40. TOTAL COST

Formula

Total Cost = Fixed Cost + Total Variable Cost

---

41. CONTRIBUTION

Easy definition

Contribution is the money remaining from sales after paying variable costs.

This money is used to:

1. Cover fixed costs
2. Generate profit

Formula

Contribution = Revenue − Variable Costs

Per unit:

Contribution per Unit = Selling Price − Variable Cost per Unit

Example

Price:

₹1,000

Variable cost:

₹600

Contribution:

₹400

---

42. CONTRIBUTION MARGIN

Easy definition

Contribution margin shows the percentage of revenue remaining after variable costs.

Formula

Contribution Margin = Contribution ÷ Revenue × 100

Example

Contribution:

₹400

Revenue:

₹1,000

Contribution margin:

40%

---

43. BREAK-EVEN POINT

Easy definition

The break-even point is the sales level at which:

Total revenue equals total cost, so profit is zero.

Formula

Break-even Units = Fixed Costs ÷ Contribution per Unit

Example

Fixed costs:

₹10 crore

Contribution per unit:

₹500

Break-even:

200,000 units

---

44. OPERATING LEVERAGE

Easy definition

Operating leverage shows how strongly profit changes when revenue changes.

Companies with:

- High fixed costs

- Low variable costs

have high operating leverage.

Example

A factory has high rent and machinery costs.

After fixed costs are covered, additional sales can create high profit.

But if sales fall, profit can decrease sharply.

---

45. ECONOMIES OF SCALE

Easy definition

Economies of scale occur when:

Producing more units reduces the average cost per unit.

Reasons:

- Fixed costs are spread across more units
- Bulk purchasing reduces prices
- Capacity is used more efficiently

---

46. DISECONOMIES OF SCALE

Easy definition

Diseconomies of scale occur when a company becomes so large that its average costs increase.

Reasons:

- Too many management layers
- Slow decisions
- Coordination problems
- Greater complexity

PART 6 — WORKING CAPITAL

47. WORKING CAPITAL

Easy definition

Working capital is the money available for a company's short-term operations.

Formula

Working Capital = Current Assets − Current Liabilities

---

48. NET WORKING CAPITAL — NWC

Easy definition

NWC measures the money tied up in normal business operations.

Formula

NWC = Accounts Receivable + Inventory – Accounts Payable

Example

Receivables:

₹100 crore

Inventory:

₹80 crore

Payables:

₹60 crore

NWC:

₹120 crore

---

49. ACCOUNTS RECEIVABLE

Easy definition

Accounts receivable is money that customers owe the company.

Example

The company sells goods worth ₹10 crore but allows customers to pay later.

Accounts receivable:

₹10 crore

---

50. ACCOUNTS PAYABLE

Easy definition

Accounts payable is money that the company owes suppliers.

Example

The company buys materials worth ₹5 crore but has not paid.

Accounts payable:

₹5 crore

---

51. INVENTORY

Easy definition

Inventory is the stock of materials or products held by a company.

Types:

- Raw materials

- Work in progress
- Finished products

---

52. CASH CONVERSION CYCLE — CCC

Easy definition

The cash conversion cycle measures:

How many days the company takes to convert cash spent on inventory back into cash received from customers.

Formula

CCC = DIO + DSO − DPO

Shorter CCC is generally better.

---

53. DIO

Full form

Days Inventory Outstanding

Easy definition

Average number of days inventory remains with the company before being sold.

Formula

DIO = Average Inventory ÷ COGS × 365

---

54. DSO

Full form

Days Sales Outstanding

Easy definition

Average number of days customers take to pay the company.

Formula

DSO = Accounts Receivable ÷ Revenue × 365

---

55. DPO

Full form

Days Payable Outstanding

Easy definition

Average number of days the company takes to pay suppliers.

Formula

DPO = Accounts Payable ÷ COGS × 365

PART 7 — TIME VALUE OF MONEY

56. TIME VALUE OF MONEY

Easy definition

Money today is worth more than the same amount in the future because money today can:

- Be invested
- Earn interest
- Be used immediately

Example

₹100 today can become ₹110 after one year at 10% interest.

Therefore:

₹100 today is worth more than ₹100 received after one year.

---

57. PRESENT VALUE — PV

Easy definition

Present value is the value today of money that will be received in the future.

Formula

PV = Future Value ÷ $(1 + r)^n$

Where:

- r = Discount rate
- n = Number of years

---

58. FUTURE VALUE — FV

Easy definition

Future value is how much today's money will become after earning a return.

Formula

FV = Present Value × $(1 + r)^n$

---

59. DISCOUNT RATE

Easy definition

The discount rate is the return required by an investor for waiting and taking risk.

It reflects:

- Time value of money
- Risk

- Opportunity cost
- Required return

Higher discount rate:

Lower present value.

PART 8 — INVESTMENT DECISIONS

60. NET PRESENT VALUE — NPV

Easy definition

NPV tells us:

How much value an investment creates today after considering future cash flows, time, and risk.

Formula

NPV = Present Value of Future Cash Flows − Initial Investment

Decision

- NPV > 0 → Creates value
- NPV = 0 → Meets the required return
- NPV < 0 → Destroys value

Example

Initial investment:

₹100 crore

Future cash flows:

₹60 crore after one year

₹60 crore after two years

Discount rate:

10%

PV of Year 1:

₹60 ÷ 1.10

= ₹54.5 crore

PV of Year 2:

₹60 ÷ 1.10^2

= ₹49.6 crore

Total PV:

₹104.1 crore

NPV:

₹104.1 − ₹100

= **₹4.1 crore**

Decision:

Accept because NPV is positive.

---

61. INTERNAL RATE OF RETURN — IRR

Easy definition

IRR is the annual return generated by an investment.

Technically:

IRR is the discount rate at which NPV equals zero.

Decision

- IRR > Required return → Accept
- IRR < Required return → Reject

Example

Investment IRR:

18%

Required return:

15%

Decision:

Accept.

---

62. RETURN ON INVESTMENT — ROI

Easy definition

ROI shows how much return is generated compared with the money invested.

Formula

ROI = Net Return ÷ Investment × 100

Example

Investment:

₹100 crore

Return:

₹20 crore

ROI:

20%

---

63. PAYBACK PERIOD

Easy definition

Payback period is the time required to recover the initial investment.

Formula

Payback Period = Initial Investment ÷ Annual Cash Flow

Example

Investment:

₹100 crore

Annual cash flow:

₹25 crore

Payback:

4 years

Limitation

It ignores:

- Time value of money
- Cash flows after payback

PART 9 — VALUATION

64. VALUATION

Easy definition

Valuation is the process of estimating:

How much a company, business, asset, or investment is worth.

Main methods:

1. Comparable-company analysis
2. Precedent transactions
3. Discounted cash flow

---

65. MARKET CAPITALIZATION

Easy definition

Market capitalization is the total market value of a company's shares.

Formula

Market Cap = Share Price × Number of Shares

Example

Share price:

₹100

Number of shares:

10 crore

Market capitalization:

₹1,000 crore

---

66. ENTERPRISE VALUE — EV

Easy definition

Enterprise value is the value of the company's entire operating business.

It considers:

- Shareholder value
- Debt
- Cash

Formula

EV = Equity Value + Debt − Cash

Example

Equity value:

₹1,000 crore

Debt:

₹300 crore

Cash:

₹100 crore

EV:

₹1,200 crore

---

67. EQUITY VALUE

Easy definition

Equity value is the value belonging to shareholders after considering debt and cash.

Formula

Equity Value = EV − Debt + Cash

---

68. EV/EBITDA

Easy definition

EV/EBITDA shows how many times a company's EBITDA investors are willing to pay for the business.

Formula

EV/EBITDA = Enterprise Value ÷ EBITDA

Example

EV:

₹1,000 crore

EBITDA:

₹100 crore

EV/EBITDA:

10×

---

69. P/E RATIO

Full form

Price-to-Earnings Ratio

Easy definition

P/E shows how much investors are willing to pay for each ₹1 of company earnings.

Formula

P/E = Market Capitalization ÷ Net Income

Example

Market value:

₹1,000 crore

Net income:

₹100 crore

P/E:

10×

---

70. DISCOUNTED CASH FLOW — DCF

Easy definition

DCF values a company by estimating the cash it will generate in the future and converting those future cash flows into today's value.

Steps

1. Forecast revenue
2. Forecast costs
3. Calculate profit
4. Calculate free cash flow
5. Discount future cash flows
6. Calculate terminal value

7. Add all present values

---

71. FREE CASH FLOW — FCF

Easy definition

Free cash flow is the cash generated by a company that is available to investors after paying operating costs and necessary investments.

Simplified formula

FCF = NOPAT + D&A − CAPEX − Increase in NWC

---

72. NOPAT

Full form

Net Operating Profit After Tax

Easy definition

NOPAT is operating profit after tax, ignoring interest.

Formula

NOPAT = EBIT × (1 − Tax Rate)

---

73. CAPITAL EXPENDITURE — CAPEX

Easy definition

CAPEX is money spent to buy, build, or improve long-term assets.

Examples

- Factory
- Machinery
- Building
- Major technology system

---

74. TERMINAL VALUE

Easy definition

Terminal value estimates the value of all cash flows expected after the detailed forecast period.

Formula

Terminal Value = Final-Year FCF × (1 + g) ÷ (WACC − g)

Where:

- g = Long-term growth rate
- WACC = Discount rate

PART 10 — RETURNS AND CAPITAL

75. RETURN ON EQUITY — ROE

Easy definition

ROE shows how efficiently the company uses shareholders' money to generate profit.

Formula

ROE = Net Income ÷ Shareholders' Equity × 100

---

76. RETURN ON ASSETS — ROA

Easy definition

ROA shows how efficiently the company uses its assets to generate profit.

Formula

ROA = Net Income ÷ Total Assets × 100

---

77. RETURN ON INVESTED CAPITAL — ROIC

Easy definition

ROIC shows how efficiently the company generates operating profit from all the capital invested in the business.

Formula

ROIC = NOPAT ÷ Invested Capital

Important

If:

ROIC > WACC

The company generally creates value.

If:

ROIC < WACC

The company may destroy value.

---

78. WACC

Full form

Weighted Average Cost of Capital

Easy definition

WACC is the average return required by the company's lenders and shareholders.

It is commonly used as a discount rate in DCF valuation.

PART 11 — DEBT AND FINANCIAL RISK

79. DEBT

Easy definition

Debt is money borrowed by a company that must be repaid, usually with interest.

Examples:

- Bank loans
- Bonds
- Long-term loans

---

80. LEVERAGE

Easy definition

Leverage means using borrowed money to finance a business or investment.

Benefit:

Can increase returns.

Risk:

Increases interest and repayment obligations.

---

81. DEBT-TO-EQUITY RATIO

Formula

Debt-to-Equity = Total Debt ÷ Equity

Meaning

Shows how much debt the company uses compared with shareholder money.

---

82. INTEREST COVERAGE RATIO

Easy definition

Shows whether the company earns enough operating profit to pay interest.

Formula

Interest Coverage = EBIT ÷ Interest Expense

Higher is generally safer.

---

83. NET DEBT

Easy definition

Net debt is the company's debt after subtracting available cash.

Formula

Net Debt = Total Debt – Cash

PART 12 — M&A FINANCE

84. ACQUISITION PREMIUM

Easy definition

Acquisition premium is the extra amount a buyer pays above the target's current value.

Formula

Premium = Purchase Price – Target Value

---

85. SYNERGY

Easy definition

Synergy means:

The combined companies create more value together than they could separately.

Types:

- Revenue synergy
- Cost synergy
- Operational synergy
- Financial synergy
- Strategic synergy

---

86. REVENUE SYNERGY

Easy definition

Revenue synergy is additional revenue created because the companies are combined.

Examples:

- Cross-selling
- New customers
- New markets
- Product bundles

---

87. COST SYNERGY

Easy definition

Cost synergy is money saved because the companies combine operations.

Examples:

- Shared offices
- Combined procurement

- Fewer duplicate employees
- Shared technology

---

88. MAXIMUM ACQUISITION PRICE

Easy definition

The maximum acquisition price is the highest price the buyer can pay while still receiving its required return.

Simplified formula

Maximum Price = Standalone Value + Synergy Value − Integration Costs − Required Buyer Return

---

89. GOODWILL

Easy definition

Goodwill is the amount paid above the fair value of the acquired company's identifiable net assets.

It may represent:

- Brand value
- Customer relationships
- Reputation
- Expected synergies

PART 13 — CUSTOMER AND BUSINESS METRICS

90. CUSTOMER ACQUISITION COST — CAC

Easy definition

CAC is the average cost required to acquire one new customer.

Formula

CAC = Sales and Marketing Cost ÷ New Customers

---

91. CUSTOMER LIFETIME VALUE — LTV

Easy definition

LTV is the estimated total profit or contribution generated by a customer during the entire relationship.

Simplified formula

LTV = Annual Customer Contribution × Customer Lifetime

---

92. LTV/CAC

Easy definition

Shows how much customer value is generated compared with acquisition cost.

Formula

LTV/CAC = LTV ÷ CAC

Higher is generally better.

---

93. CHURN RATE

Easy definition

Churn rate is the percentage of customers who stop using the company during a period.

Formula

Churn = Customers Lost ÷ Starting Customers × 100

---

94. RETENTION RATE

Easy definition

Retention rate is the percentage of customers who continue using the company.

Formula

Retention = 100% − Churn

---

95. MARKET SHARE

Easy definition

Market share is the percentage of total market sales controlled by a company.

Formula

Market Share = Company Sales ÷ Total Market Sales × 100

---

96. GROWTH RATE

Easy definition

Growth rate shows how much a value increased or decreased compared with the previous period.

Formula

Growth Rate = (Current Value − Previous Value) ÷ Previous Value × 100

---

97. CAGR

Full form

Compound Annual Growth Rate

Easy definition

CAGR is the average annual growth rate over several years, assuming growth compounds.

Formula

CAGR = (Ending Value ÷ Beginning Value)^(1/n) – 1

FINAL FORMULA SHEET

Revenue

= Price × Quantity

Gross Profit

= Revenue − COGS

Gross Margin

= Gross Profit ÷ Revenue

EBITDA

= Revenue − COGS − OPEX

EBIT

= EBITDA − Depreciation − Amortization

PBT

= EBIT − Interest

Net Income

= PBT − Tax

Total Cost

= Fixed Cost + Variable Cost

Contribution

= Revenue − Variable Cost

Break-even Units

= Fixed Cost ÷ Contribution per Unit

Working Capital

= Current Assets − Current Liabilities

NWC

= Receivables + Inventory − Payables

Cash Conversion Cycle

= DIO + DSO − DPO

Future Value

$= PV \times (1 + r)^n$

Present Value

$= FV \div (1 + r)^n$

NPV

= PV of Future Cash Flows − Initial Investment

ROI

= Return ÷ Investment

Payback

= Investment ÷ Annual Cash Flow

NOPAT

= EBIT × (1 − Tax Rate)

Free Cash Flow

= NOPAT + D&A − CAPEX − Increase in NWC

Enterprise Value

= Equity Value + Debt − Cash

Equity Value

= EV − Debt + Cash

ROIC

= NOPAT ÷ Invested Capital

Market Share

= Company Sales ÷ Market Sales

Growth Rate

= Change ÷ Previous Value

FINAL MBB FINANCE FLOW

Revenue

↓

COGS

↓

Gross Profit

↓

Operating Expenses

↓

EBITDA

↓

Depreciation and Amortization

↓

EBIT

↓

Interest

↓

Profit Before Tax

↓

Tax

↓

Net Income

↓

Adjust for:

D&A

CAPEX

Working Capital

↓

Free Cash Flow

↓

Discount Future Cash Flows

↓

NPV / Company Value

↓

Investment Decision

PART 3 — ADVANCED FINANCE + ACCOUNTING FOR MBB CASE INTERVIEWS

98. CAPITAL STRUCTURE

Easy definition

Capital structure is the **mix of debt and equity a company uses to finance its business**.

Main sources

- Equity
- Debt
- Sometimes preferred stock / other hybrid instruments

Example

A company needs ₹100 crore.

It could use:

- ₹60 crore equity
- ₹40 crore debt

Capital structure = **60% equity + 40% debt**

MBB use

Useful in:

- Financing decisions
- M&A
- Leveraged buyouts
- Financial-risk analysis
- Cost of capital

99. COST OF DEBT

Easy definition

Cost of debt is the effective cost a company pays for borrowed money.

Example

Debt = ₹100 crore
Interest rate = 10%

Annual interest =

₹100 × 10% = ₹10 crore

So the pre-tax cost of debt is approximately **10%**.

After-tax cost of debt

Interest expense generally provides a tax benefit.

After-tax Cost of Debt = Pre-tax Cost of Debt × (1 − Tax Rate)

Example:

Cost of debt = 10%
Tax rate = 30%

After-tax cost:

10% × (1 − 30%) = 7%

---

100. COST OF EQUITY

Easy definition

Cost of equity is the **return shareholders require for investing in the company**.

Unlike debt, equity does not have a fixed contractual interest payment.

Common formula — CAPM

Cost of Equity = Risk-Free Rate + Beta × Market Risk Premium

Where:

- Risk-free rate = return on a relatively risk-free investment
- Beta = sensitivity of the company's returns to the market
- Market risk premium = expected market return − risk-free rate

Example

Risk-free rate = 7%
Beta = 1.2
Market risk premium = 6%

Cost of equity:

7% + 1.2 × 6% = 14.2%

---

101. WACC — DETAILED

WACC = Weighted Average Cost of Capital

It represents the average required return from the company's:

- Debt holders
- Equity holders

Formula

WACC = (E/V × Cost of Equity) + (D/V × Cost of Debt × (1 − Tax Rate))

Where:

- E = Equity
- D = Debt
- V = Debt + Equity

Example

Equity = ₹60 crore
Debt = ₹40 crore

Therefore:

V = ₹100 crore

Cost of equity = 14%
Pre-tax cost of debt = 8%
Tax = 25%

WACC:

= 60% × 14% + 40% × 8% × 75%

= 8.4% + 2.4%

= **10.8%**

MBB application

WACC is especially important in:

- DCF
- Investment decisions
- Acquisition valuation
- Comparing investment returns with required returns

---

102. RISK-FREE RATE

Easy definition

The risk-free rate is the return expected from an investment considered to have extremely low default risk.

It is commonly represented using government securities.

MBB relevance

It is an input in:

- CAPM
- Cost of equity
- Valuation

---

103. BETA

Easy definition

Beta measures how sensitive an investment's returns are to movements in the overall market.

Interpretation

Beta = 1

Moves approximately with the market.

Beta > 1

More sensitive than the market.

Beta < 1

Less sensitive than the market.

Example

Beta = 1.5

If the market rises by 10%, the company's stock might be expected to move approximately 15%, all else equal.

Important: Beta is a statistical measure, not a guaranteed prediction.

---

104. MARKET RISK PREMIUM

Formula

Market Risk Premium = Expected Market Return – Risk-Free Rate

Example

Expected market return = 13%

Risk-free rate = 7%

Market risk premium:

13% – 7% = 6%

---

105. FINANCIAL LEVERAGE

Easy definition

Financial leverage means using **debt to finance the company**.

Benefit

Debt can increase shareholders' returns when the business earns more than the effective cost of debt.

Risk

Debt creates:

- Interest obligations
- Repayment obligations
- Financial risk

Example

A company earns a strong operating return but has significant debt.

A portion of the operating earnings must be used to pay interest.

Therefore, debt can amplify both **returns and downside risk** for equity holders.

---

106. DEBT SERVICE

Easy definition

Debt service is the cash required to meet debt obligations.

It can include:

- Interest payments
- Principal repayment

MBB application

Important when assessing:

- Financial health
- Debt capacity
- Acquisition financing
- Distressed companies

---

107. DSCR

Full form

Debt Service Coverage Ratio

Easy definition

DSCR measures whether a company generates enough cash to cover its debt-service obligations.

A simplified formula is:

DSCR = Cash Available for Debt Service ÷ Debt Service

Example

Cash available = ₹30 crore

Debt service = ₹20 crore

DSCR:

30 ÷ 20 = 1.5×

A higher ratio generally indicates greater debt-servicing capacity.

---

108. LIQUIDITY

Easy definition

Liquidity is the company's ability to meet its **short-term financial obligations**.

Examples

A company with sufficient:

- Cash
- Receivables
- Other current assets

may have stronger short-term liquidity.

Important distinction

Liquidity ≠ Profitability

A profitable company can still experience a cash shortage.

---

109. CURRENT RATIO

Formula

Current Ratio = Current Assets ÷ Current Liabilities

Example

Current assets = ₹200 crore

Current liabilities = ₹100 crore

Current ratio:

2.0×

Meaning

The company has ₹2 of current assets for every ₹1 of current liabilities.

---

110. QUICK RATIO

Easy definition

The quick ratio measures short-term liquidity using assets that can generally be converted to cash relatively quickly.

A common formula is:

Quick Ratio = (Cash + Marketable Securities + Accounts Receivable) ÷ Current Liabilities

Why exclude inventory?

Inventory may take time to sell and convert into cash.

---

111. SOLVENCY

Easy definition

Solvency refers to a company's ability to meet its **long-term financial obligations**.

Liquidity vs Solvency

Liquidity → Short term

Can the company pay upcoming obligations?

Solvency → Long term

Can the company sustain and repay its long-term obligations?

---

112. GROSS MARGIN VS EBITDA MARGIN VS EBIT MARGIN

Gross Margin

Gross Profit ÷ Revenue

Shows profitability after direct costs.

EBITDA Margin

EBITDA ÷ Revenue

Shows operating profitability before D&A.

EBIT Margin

EBIT ÷ Revenue

Shows operating profitability after D&A.

Example

Revenue = ₹100 crore

Gross profit = ₹40 crore

EBITDA = ₹25 crore

EBIT = ₹20 crore

Therefore:

Gross margin = **40%**

EBITDA margin = **25%**

EBIT margin = **20%**

MBB use

Margins help identify **where profitability is being lost**.

---

113. MARGIN WATERFALL

A useful way to analyze profitability:

Revenue

↓ – COGS

Gross Profit

↓ – OPEX

EBITDA

↓ – D&A

EBIT

↓ – Interest

PBT

↓ – Tax

Net Income

MBB application

If net margin falls, work backward:

Net income ↓

→ Is tax higher?

→ Is interest higher?

→ Is EBIT lower?

→ Is EBITDA lower?

→ Is gross margin lower?

→ Did price fall, volume fall, or costs increase?

This is a common profitability-case approach.

---

114. PRICE-VOLUME-MIX ANALYSIS

Easy definition

Revenue changes can often be decomposed into:

- Price effect
- Volume effect
- Product-mix effect

Example

Last year:

10,000 units × ₹100

Revenue = ₹10 lakh

This year:

12,000 units × ₹110

Revenue = ₹13.2 lakh

Revenue increased because of both:

- Higher volume
- Higher price

Why mix matters

Suppose a company sells:

- Product A: high margin
- Product B: low margin

Even if total revenue remains stable, shifting sales toward Product B can reduce profitability.

MBB use

Extremely useful in:

- Profitability cases
- Revenue decline
- Pricing cases
- Commercial due diligence

---

115. PRICE ELASTICITY OF DEMAND

Easy definition

Price elasticity measures how strongly demand responds to a change in price.

Formula

Price Elasticity = % Change in Quantity Demanded ÷ % Change in Price

Example

Price increases by 10%.

Quantity demanded falls by 20%.

Elasticity:

−20% ÷ 10% = −2

The negative sign indicates that demand generally falls when price rises.

MBB application

Used in:

- Pricing strategy
- Revenue optimization
- Market analysis

---

116. UNIT ECONOMICS

Easy definition

Unit economics examines the **revenue and costs associated with one unit of business activity**, such as one customer, order, or product.

Example

One customer generates:

Revenue = ₹1,000

Variable cost = ₹600

Contribution = ₹400

If customer acquisition cost = ₹300:

Contribution after acquisition cost =

₹400 – ₹300 = ₹100

MBB use

Important for:

- Startups
- Digital businesses
- Subscription businesses
- E-commerce
- Market-entry cases

---

117. CUSTOMER ACQUISITION PAYBACK

Easy definition

Customer acquisition payback measures how long it takes to recover CAC from customer contribution.

Simplified formula

CAC Payback Period = CAC ÷ Monthly Customer Contribution

Example

CAC = ₹600

Monthly contribution = ₹100

Payback:

6 months

Shorter payback is generally attractive because cash is recovered faster.

---

118. RECURRING REVENUE

Easy definition

Recurring revenue is revenue expected to repeat regularly.

Examples:

- Monthly subscriptions
- Annual software contracts
- Recurring service contracts

MBB relevance

Recurring revenue can provide:

- Greater predictability
- Higher customer lifetime value
- More stable cash generation

---

119. ARR

Full form

Annual Recurring Revenue

Easy definition

ARR is the annualized value of recurring revenue.

Example

A software company has:

1,000 customers

Each pays ₹12,000 per year.

ARR:

1,000 × ₹12,000 = ₹1.2 crore

---

120. MRR

Full form

Monthly Recurring Revenue

Easy definition

MRR measures recurring revenue generated in a month.

Formula

MRR = Number of Paying Customers × Average Monthly Recurring Revenue per Customer

Relationship

Approximately:

ARR = MRR × 12

when recurring revenue remains consistent.

---

121. EBITDA VS CASH FLOW

Important MBB distinction

EBITDA is **not cash flow**.

EBITDA excludes:

- Interest
- Tax
- D&A

But a company may still need cash for:

- CAPEX
- Working capital
- Interest
- Taxes

Therefore:

High EBITDA ≠ High Free Cash Flow necessarily

---

122. EBITDA TO FREE CASH FLOW BRIDGE

A simplified bridge:

EBITDA

↓ – Cash taxes

↓ – Interest, where relevant to the cash-flow definition

↓ – CAPEX

↓ – Increase in NWC

Free Cash Flow

Depending on the specific FCF definition, interest may be handled differently.

For **unlevered FCF**, the standard DCF formulation typically starts from EBIT/NOPAT rather than deducting interest.

---

123. CAPEX VS OPEX

CAPEX

Money spent on **long-term assets**.

Examples:

- Factory
- Machinery
- Major equipment

CAPEX generally appears on the balance sheet initially and is depreciated over time.

OPEX

Day-to-day operating expenses.

Examples:

- Salaries
- Rent
- Advertising
- Utilities

OPEX is generally expensed in the period incurred.

MBB application

A company may improve short-term EBITDA by reducing OPEX, but delaying necessary CAPEX can hurt future operations.

---

124. DEPRECIATION VS CAPEX

This is a very common interview concept.

CAPEX = actual investment/spending

Depreciation = accounting allocation of that investment over time

Example

Company purchases machinery for ₹100 crore.

Immediately:

CAPEX = ₹100 crore

If useful life = 10 years:

Annual straight-line depreciation:

₹10 crore

The company does **not** pay ₹10 crore cash every year because of depreciation.

---

125. DEPRECIATION AND TAX

Depreciation is generally a non-cash accounting expense but can reduce taxable income.

Example

EBIT before depreciation = ₹30 crore

Depreciation = ₹10 crore

Taxable operating profit may be reduced to:

₹20 crore

At a 30% tax rate:

Tax = ₹6 crore

Without depreciation:

Tax = ₹9 crore

Tax saving:

₹3 crore

This is called a **depreciation tax shield**.

---

126. TAX SHIELD

Easy definition

A tax shield is a reduction in taxable income caused by an allowable expense.

Common examples include:

- Interest expense
- Depreciation

Interest tax shield

If interest = ₹10 crore

Tax rate = 30%

Potential tax saving:

₹10 × 30% = ₹3 crore

---

127. SENSITIVITY ANALYSIS

Easy definition

Sensitivity analysis asks:

"What happens to the result if one important assumption changes?"

Example

DCF valuation:

Base value = ₹1,000 crore

If WACC increases:

Value may decrease.

If long-term growth increases:

Value may increase.

MBB use

Used to identify:

- Key assumptions
- Risks
- Upside/downside
- Decision robustness

---

128. SCENARIO ANALYSIS

Easy definition

Scenario analysis evaluates different combinations of assumptions.

Typical scenarios:

Base case

Most likely assumptions.

Upside case

Favorable assumptions.

Downside case

Unfavorable assumptions.

Example

Scenario	Revenue Growth	EBITDA Margin
Downside	3%	15%
Base	7%	20%
Upside	12%	24%

MBB use

Helps management understand the range of possible outcomes.

---

129. BREAKEVEN REVENUE

Earlier we calculated break-even in units.

We can also calculate break-even revenue.

Break-even Revenue = Fixed Costs ÷ Contribution Margin

Example

Fixed costs = ₹20 crore

Contribution margin = 40%

Break-even revenue:

₹20 ÷ 40% = ₹50 crore

At ₹50 crore revenue, the company breaks even.

---

130. MARGIN OF SAFETY

Easy definition

Margin of safety shows how far actual/current sales are above the break-even level.

Formula

Margin of Safety = Actual Sales – Break-even Sales

As a percentage:

Margin of Safety % = (Actual Sales − Break-even Sales) ÷ Actual Sales × 100

Example

Actual sales = ₹80 crore

Break-even sales = ₹50 crore

Margin of safety:

₹30 crore

Percentage:

37.5%

---

131. OPERATING LEVERAGE — FORMULA

A simplified degree of operating leverage:

DOL = Contribution ÷ EBIT

Example

Contribution = ₹40 crore

EBIT = ₹20 crore

DOL:

2×

Approximately, a 1% change in sales can produce a 2% change in EBIT, assuming the relevant operating assumptions remain constant.

---

132. FINANCIAL LEVERAGE — CONCEPT

Financial leverage measures how debt magnifies the impact of operating performance on shareholders' earnings.

Example

If EBIT increases but interest expense remains relatively fixed, the increase can flow disproportionately to profit available to equity holders.

However, the reverse is also true:

Lower EBIT → greater pressure from fixed interest obligations.

---

133. DUPONT ANALYSIS

Easy definition

DuPont analysis breaks ROE into several components to understand **why ROE is high or low**.

Formula

ROE = Net Profit Margin × Asset Turnover × Financial Leverage

Where:

Net Profit Margin = Net Income ÷ Revenue

Asset Turnover = Revenue ÷ Assets

Financial Leverage = Assets ÷ Equity

Why useful?

A company can have high ROE because of:

- High profit margins
- Efficient use of assets
- High financial leverage

MBB application

Useful for diagnosing company performance rather than looking only at ROE.

---

134. EV VS EQUITY VALUE — CRITICAL

Enterprise Value

Value of the operating business attributable to both debt and equity providers.

EV = Equity Value + Debt − Cash

Equity Value

Value attributable to shareholders.

Equity Value = EV − Debt + Cash

Example

EV = ₹1,000 crore

Debt = ₹300 crore

Cash = ₹100 crore

Equity value:

₹1,000 − ₹300 + ₹100 = ₹800 crore

---

135. WHY EV/EBITDA BUT P/E?

This is an important valuation distinction.

EV/EBITDA

Uses:

Enterprise Value / EBITDA

Because EV represents value to both debt and equity holders, it is paired with an operating metric before interest.

P/E

Uses:

Equity Value / Net Income

Both are equity-level measures.

Easy rule

EV ↔ Operating earnings

Equity Value ↔ Equity earnings

---

136. TRADING COMPARABLES

Easy definition

Comparable-company analysis values a company using valuation multiples of similar publicly traded companies.

Steps

1. Identify comparable companies.
2. Collect financial metrics.
3. Calculate valuation multiples.
4. Select an appropriate multiple.
5. Apply the multiple to the target company.

Example

Comparable companies trade at:

10× EV/EBITDA

Target EBITDA:

₹100 crore

Implied EV:

10 × ₹100 = ₹1,000 crore

---

137. PRECEDENT TRANSACTIONS

Easy definition

Precedent transaction analysis values a company using multiples paid in **past acquisitions of similar companies**.

Important

Transaction multiples can be higher than public-market trading multiples because an acquirer may pay a:

- Control premium
- Strategic premium
- Synergy value

MBB use

Common in:

- M&A
- Acquisition valuation

- Deal analysis

---

138. DCF — COMPLETE LOGIC

The DCF process can be remembered as:

Revenue

↓

EBITDA

↓

EBIT

↓

NOPAT

↓

+ D&A

↓

– CAPEX

↓

– Increase in NWC

↓

Free Cash Flow

↓

Discount FCF

↓

Terminal Value

↓

Discount Terminal Value

↓

Enterprise Value

Core idea

Future cash → convert to today's value

---

139. DCF TERMINAL VALUE — TWO METHODS

Method 1: Perpetuity Growth Method

$TV = FCF_n \times (1 + g) \div (WACC - g)$

Where:

- FCF_n = final forecast-year FCF
- g = perpetual growth rate

Method 2: Exit Multiple Method

TV = Final-Year EBITDA × Exit EV/EBITDA Multiple

MBB application

Both methods estimate value beyond the explicit forecast period.

---

140. DCF — IMPORTANT CHECK

For the perpetuity-growth method:

WACC must be greater than g.

Otherwise the formula does not produce a meaningful finite terminal value.

---

141. ACCRETION AND DILUTION

Easy definition

Accretion/dilution analysis asks whether an acquisition increases or decreases the buyer's earnings per share.

Accretive

Post-deal EPS > Pre-deal EPS

Dilutive

Post-deal EPS < Pre-deal EPS

MBB/M&A use

Used to evaluate the financial impact of a proposed acquisition.

---

142. M&A DEAL LOGIC

A simplified acquisition analysis:

Standalone Target Value

-

Synergy Value

–

Integration Costs

=

Maximum Economic Value of Deal

Then compare:

Value Created – Purchase Price

If positive:

→ Potentially value-creating.

If negative:

→ Potentially value-destroying.

---

143. SYNERGY REALIZATION

Not all announced synergies necessarily become actual savings or revenue.

Consider:

- Size of synergy
- Timing
- Probability of realization
- Integration cost
- Execution risk

MBB application

A consultant should ask:

“What is the realistic, achievable synergy—not just the theoretical synergy?”

---

144. ORGANIC VS INORGANIC GROWTH

Organic growth

Growth generated from the company's existing business.

Examples:

- Selling more products
- Increasing prices
- Entering new customer segments

Inorganic growth

Growth through external transactions.

Examples:

- Acquisitions
- Mergers

---

145. MARKET GROWTH VS MARKET SHARE

A company's revenue can grow for two very different reasons:

Market growth

The entire market becomes larger.

Market-share gain

The company captures a larger portion of the existing market.

Example

Market:

₹1,000 crore → ₹1,200 crore

Company:

₹100 crore → ₹132 crore

Market growth:

20%

Company growth:

32%

Market share:

10% → 11%

The company grew faster than the market because it gained share.

---

146. GROWTH DECOMPOSITION

Revenue can often be decomposed into:

Revenue = Price × Volume

Therefore growth can come from:

- Price increases
- Volume increases
- Product/customer mix changes

For a market:

Company Revenue = Market Size × Market Share

Therefore company growth can come from:

- Market growth
- Market-share gain

This is extremely useful in MBB growth cases.

---

147. INFLATION IMPACT ON FINANCIALS

Inflation can affect:

Revenue

Prices may increase.

COGS

Raw materials may become more expensive.

OPEX

Wages, rent, utilities and other expenses may rise.

Interest

Borrowing costs may rise when interest rates increase.

MBB approach

Don't simply ask:

“Did revenue increase?”

Ask:

“Did real profitability improve after considering cost inflation?”

---

148. RECESSION IMPACT

During an economic slowdown:

Potential effects include:

- Lower demand
- Lower volumes
- Higher defaults
- Working-capital pressure
- Lower margins
- Reduced investment
- Greater financing risk

Companies with high fixed costs or high debt may be particularly vulnerable.

---

149. COMMON-SIZE INCOME STATEMENT

Easy definition

A common-size income statement expresses every item as a percentage of revenue.

Example

Revenue = ₹100 crore

COGS = ₹60 crore

OPEX = ₹15 crore

Therefore:

COGS = **60% of revenue**

OPEX = **15%**

EBITDA = **25%**

MBB use

Useful for comparing companies of different sizes.

---

150. COMMON-SIZE BALANCE SHEET

Each balance-sheet item can be expressed as a percentage of total assets.

Example

Total assets = ₹500 crore

Inventory = ₹100 crore

Inventory as % of assets:

100 ÷ 500 = 20%

Useful for comparing financial structures across companies.

---

151. YEAR-OVER-YEAR — YoY

Easy definition

Compares a metric with the same period in the previous year.

Formula

YoY Growth = (Current Year − Previous Year) ÷ Previous Year × 100

Example

Revenue:

2025 = ₹100 crore

2026 = ₹120 crore

YoY growth:

20%

---

152. QUARTER-OVER-QUARTER — QoQ

Easy definition

Compares performance with the immediately preceding quarter.

Example:

Q1 revenue = ₹100 crore

Q2 revenue = ₹110 crore

QoQ growth:

10%

---

153. CAGR VS AVERAGE GROWTH

Suppose revenue increases:

₹100 → ₹120 → ₹150

CAGR measures the compounded annual growth rate between the beginning and ending values.

It is different from simply taking the arithmetic average of annual growth rates.

Remember

CAGR = compounded growth

not

simple average growth

---

154. FINANCIAL STATEMENT LINKAGES

This is **very important for MBB interviews**.

Income Statement → Balance Sheet

Net income contributes to retained earnings.

Depreciation reduces the carrying value of fixed assets over time.

Income Statement → Cash Flow

Net income is a starting point in the indirect cash-flow method.

Non-cash expenses such as depreciation are added back.

Balance Sheet → Cash Flow

Changes in:

- Receivables
- Inventory
- Payables

affect operating cash flow.

Core relationship

Income Statement = Performance

Balance Sheet = Position

Cash Flow Statement = Cash movement

---

155. CLASSIC MBB QUESTION: PROFIT UP, CASH DOWN

Suppose:

Net income increases

but

Cash decreases.

Possible reasons:

- Accounts receivable increased

- Inventory increased
- CAPEX increased
- Debt repayment increased
- Dividends increased

Example

Company records ₹20 crore additional sales on credit.

Profit can increase.

But until customers pay:

Cash has not increased by ₹20 crore.

Therefore:

Profit ↑ but cash ↓/lower than expected

can happen.

---

156. CLASSIC MBB QUESTION: REVENUE UP, PROFIT DOWN

Possible reasons:

Revenue side

- Price decreased
- Volume mix deteriorated
- Low-margin products increased

Cost side

- Raw-material prices increased
- Labor costs increased
- Logistics costs increased
- Marketing costs increased

MBB framework

Profit = Revenue – Costs

So investigate:

Revenue

→ Price

→ Volume

→ Mix

and

Costs

→ Variable

→ Fixed

→ COGS

→ OPEX

---

157. CLASSIC MBB QUESTION: PROFIT DOWN, WHAT DO YOU CHECK?

Use this sequence:

1. Revenue

Price × Volume × Mix

2. COGS

- Raw materials
- Labor
- Manufacturing
- Logistics

3. OPEX

- Sales
- Marketing
- Administration
- R&D

4. D&A

Has depreciation/amortization changed?

5. Interest

Has debt or interest rates increased?

6. Tax

Has the effective tax burden changed?

This gives a structured profitability diagnosis.

---

158. PROFITABILITY TREE

A highly useful MBB structure:

Profit

=

Revenue – Cost

Revenue

Price × Volume

Then investigate:

- Customers
- Products
- Geography
- Channels
- Market share

Cost

Fixed + Variable

Then investigate:

- Raw materials
- Labor
- Logistics
- Manufacturing
- SG&A
- Marketing

This is more important for MBB cases than memorizing dozens of finance formulas.

---

159. INVESTMENT DECISION FRAMEWORK

When evaluating an investment:

Step 1 — Initial investment

How much cash is required?

Step 2 — Future cash flows

How much cash will the investment generate?

Step 3 — Risk

How uncertain are those cash flows?

Step 4 — Discount rate

What return is required?

Step 5 — NPV

Does the investment create value?

Step 6 — Sensitivity

What happens if assumptions change?

Step 7 — Strategic factors

Does the investment provide additional strategic benefits?

---

160. MBB FINANCE MASTER FLOW

For almost any finance-heavy case, think:

Revenue

↓

Price × Volume × Mix

↓

COGS

↓

Gross Profit

↓

OPEX

↓

EBITDA

↓

D&A

↓

EBIT

↓

Interest + Tax

↓

Net Income

↓

Working Capital + CAPEX

↓

Free Cash Flow

↓

Return / NPV / Valuation

↓

Investment Decision

---

161. FINAL HIGH-PRIORITY MBB FORMULA SHEET

Profitability

Revenue = Price × Volume

Gross Profit = Revenue − COGS

Gross Margin = Gross Profit ÷ Revenue

EBITDA = Revenue – COGS – OPEX

EBITDA Margin = EBITDA ÷ Revenue

EBIT = EBITDA – D&A

EBIT Margin = EBIT ÷ Revenue

PBT = EBIT – Interest

Net Income = PBT – Tax

Cost

Total Cost = Fixed Cost + Variable Cost

Contribution = Revenue – Variable Cost

Contribution per Unit = Price – Variable Cost per Unit

Contribution Margin = Contribution ÷ Revenue

Break-even Units = Fixed Costs ÷ Contribution per Unit

Break-even Revenue = Fixed Costs ÷ Contribution Margin

Working Capital

NWC = Receivables + Inventory – Payables

CCC = DIO + DSO – DPO

DIO = Inventory ÷ COGS × 365

DSO = Receivables ÷ Revenue × 365

DPO = Payables ÷ COGS × 365

Returns

ROI = Net Return ÷ Investment

ROE = Net Income ÷ Equity

ROA = Net Income ÷ Assets

ROIC = NOPAT ÷ Invested Capital

NOPAT = EBIT × (1 – Tax Rate)

Valuation

EV = Equity Value + Debt – Cash

Equity Value = EV – Debt + Cash

EV/EBITDA = EV ÷ EBITDA

P/E = Market Capitalization ÷ Net Income

Market Cap = Share Price × Shares Outstanding

DCF

PV = FV ÷ $(1+r)^n$

FV = PV × $(1+r)^n$

NPV = PV of Future Cash Flows − Initial Investment

FCF = NOPAT + D&A − CAPEX − Increase in NWC

TV = FCF_n × (1+g) ÷ (WACC−g)

Capital

WACC = (E/V × Ke) + (D/V × Kd × (1−Tax))

Ke = Rf + Beta × Market Risk Premium

Market Risk Premium = Market Return − Risk-Free Rate

Growth

Growth = (Current − Previous) ÷ Previous

CAGR = (Ending ÷ Beginning)^(1/n) − 1

Market Share = Company Sales ÷ Market Sales

Customer Economics

CAC = Sales & Marketing Cost ÷ New Customers

LTV = Customer Contribution × Customer Lifetime

LTV/CAC = LTV ÷ CAC

Churn = Customers Lost ÷ Starting Customers

Retention = 1 − Churn

---

162. WHAT YOU SHOULD BE ABLE TO DO FOR MBB

After Parts 1–3, you should be able to look at a case and quickly answer:

Profitability

“Why is profit falling?”

→ Revenue / COGS / OPEX / D&A / interest / tax

Revenue

“How can we increase revenue?”

→ Price / volume / mix / customers / market / share

Cost

“How can we reduce costs?”

→ Fixed / variable / procurement / labor / logistics / operations

Investment

“Should we invest ₹500 crore?”

→ Cash flows / NPV / IRR / payback / risk / strategic value

M&A

“Should we acquire this company?”

→ Standalone value / purchase price / synergies / integration costs / financing / returns

Valuation

“What is this company worth?”

→ Trading comps / precedents / DCF

Cash

“Why is cash falling despite profit?”

→ Working capital / CAPEX / financing / taxes / interest

Growth

“How do we grow?”

→ Market growth / market share / price / volume / new products / new geographies / M&A

MBB PRIORITY

If your objective is **MBB case interviews**, don't spend equal time on every formula.

🔴 **MUST MASTER**

- Revenue = Price × Volume
- Fixed vs variable costs
- Contribution
- Break-even
- Gross margin
- EBITDA / EBITDA margin
- Profitability tree
- Price-volume-mix
- Market share
- Growth / CAGR
- Working capital
- Cash flow

- Three financial statements
- NPV
- ROI
- EV vs Equity Value
- Synergies
- Basic valuation

🔴 **SHOULD KNOW WELL**

- DCF
- FCF
- WACC
- ROIC
- ROE / ROA
- EV/EBITDA
- P/E
- CAPEX
- D&A
- Leverage
- LTV/CAC
- Sensitivity / scenario analysis

🟡 **KNOW CONCEPTUALLY**

- CAPM
- Beta
- DSCR
- Current ratio
- Quick ratio
- DuPont
- Accretion/dilution
- Precedent transactions
- Tax shield
- Common-size statements
- ARR / MRR

The key for MBB is not merely knowing these definitions. You need to be able to use the numbers quickly inside a case and explain what the result means for the business.